WORKED SOLUTION

STEP II 1987, Question 9

Pure • Orthogonal Matrices

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q9

The key idea

The condition $\mathbf{B}^{\mathrm{T}}\mathbf{B} = \mathbf{I}$ has inverses on both outside ends once we transpose $\mathbf{B}$. Because those inverse matrices are invertible, we can multiply through by $\mathbf{I} - \mathbf{A}^{\mathrm{T}}$ on the left and $\mathbf{I} - \mathbf{A}$ on the right without losing information. What's left is a polynomial identity in $\mathbf{A}$ and $\mathbf{A}^{\mathrm{T}}$ that collapses almost at once. The thing to be careful about is that every step must be reversible, since we need "if and only if".

Solution

The standard results we use are:

- $(\mathbf{XY})^{\mathrm{T}} = \mathbf{Y}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}$, and transposition is linear, so $(\mathbf{I} \pm \mathbf{A})^{\mathrm{T}} = \mathbf{I} \pm \mathbf{A}^{\mathrm{T}}$ (as $\mathbf{I}^{\mathrm{T}} = \mathbf{I}$);
- if $\mathbf{X}$ is non-singular then so is $\mathbf{X}^{\mathrm{T}}$, and $(\mathbf{X}^{-1})^{\mathrm{T}} = (\mathbf{X}^{\mathrm{T}})^{-1}$.

Step 1: the transpose of B. Since $\mathbf{I} - \mathbf{A}$ is non-singular, so is its transpose $\mathbf{I} - \mathbf{A}^{\mathrm{T}}$. Reversing the order of the product,

$$\mathbf{B}^{\mathrm{T}} = \left((\mathbf{I} - \mathbf{A})^{-1}\right)^{\mathrm{T}}(\mathbf{I} + \mathbf{A})^{\mathrm{T}} = (\mathbf{I} - \mathbf{A}^{\mathrm{T}})^{-1}(\mathbf{I} + \mathbf{A}^{\mathrm{T}}).$$

Hence

$$\mathbf{B}^{\mathrm{T}}\mathbf{B} = (\mathbf{I} - \mathbf{A}^{\mathrm{T}})^{-1}\,(\mathbf{I} + \mathbf{A}^{\mathrm{T}})(\mathbf{I} + \mathbf{A})\,(\mathbf{I} - \mathbf{A})^{-1}.$$

Step 2: clear the inverses, reversibly. Write $\mathbf{P} = \mathbf{I} - \mathbf{A}^{\mathrm{T}}$, $\mathbf{Q} = \mathbf{I} - \mathbf{A}$ and $\mathbf{M} = (\mathbf{I} + \mathbf{A}^{\mathrm{T}})(\mathbf{I} + \mathbf{A})$, so that $\mathbf{B}^{\mathrm{T}}\mathbf{B} = \mathbf{P}^{-1}\mathbf{M}\mathbf{Q}^{-1}$. We claim

$$\mathbf{P}^{-1}\mathbf{M}\mathbf{Q}^{-1} = \mathbf{I} \quad \Longleftrightarrow \quad \mathbf{M} = \mathbf{PQ}.$$

($\Rightarrow$) Multiply on the left by $\mathbf{P}$ and on the right by $\mathbf{Q}$: $\mathbf{M} = \mathbf{PIQ} = \mathbf{PQ}$.
($\Leftarrow$) Multiply on the left by $\mathbf{P}^{-1}$ and on the right by $\mathbf{Q}^{-1}$: $\mathbf{P}^{-1}\mathbf{M}\mathbf{Q}^{-1} = \mathbf{P}^{-1}\mathbf{PQQ}^{-1} = \mathbf{I}$.

Ask yourself Why can't we just "multiply both sides" and stop there?

Multiplying an equation by a matrix always gives a true equation, but in general it can't be undone: from $\mathbf{XY} = \mathbf{XZ}$ we can't conclude $\mathbf{Y} = \mathbf{Z}$ unless $\mathbf{X}$ is invertible. Here the reverse direction works only because $\mathbf{P}$ and $\mathbf{Q}$ have inverses, which is exactly where the hypothesis that $\mathbf{I} - \mathbf{A}$ is non-singular (and hence so is $\mathbf{I} - \mathbf{A}^{\mathrm{T}}$) is used. Note too that we must keep $\mathbf{P}$ on the left and $\mathbf{Q}$ on the right, as matrix multiplication isn't commutative.

Step 3: expand. Expanding both sides of $\mathbf{M} = \mathbf{PQ}$, carefully keeping the order of factors,

$$(\mathbf{I} + \mathbf{A}^{\mathrm{T}})(\mathbf{I} + \mathbf{A}) = \mathbf{I} + \mathbf{A} + \mathbf{A}^{\mathrm{T}} + \mathbf{A}^{\mathrm{T}}\mathbf{A},$$
$$(\mathbf{I} - \mathbf{A}^{\mathrm{T}})(\mathbf{I} - \mathbf{A}) = \mathbf{I} - \mathbf{A} - \mathbf{A}^{\mathrm{T}} + \mathbf{A}^{\mathrm{T}}\mathbf{A}.$$

The $\mathbf{I}$ and $\mathbf{A}^{\mathrm{T}}\mathbf{A}$ terms appear on both sides, so subtracting,

$$\mathbf{M} = \mathbf{PQ} \iff 2(\mathbf{A} + \mathbf{A}^{\mathrm{T}}) = \mathbf{O} \iff \mathbf{A} + \mathbf{A}^{\mathrm{T}} = \mathbf{O}.$$

Each of these is reversible: adding or subtracting the same matrix on both sides, and multiplying by the non-zero scalar 2 or $\frac{1}{2}$.

Conclusion. Chaining Steps 1–3, every link being an equivalence,

$$\mathbf{B}^{\mathrm{T}}\mathbf{B} = \mathbf{I} \iff (\mathbf{I} + \mathbf{A}^{\mathrm{T}})(\mathbf{I} + \mathbf{A}) = (\mathbf{I} - \mathbf{A}^{\mathrm{T}})(\mathbf{I} - \mathbf{A}) \iff \mathbf{A} + \mathbf{A}^{\mathrm{T}} = \mathbf{O},$$

which is what we had to prove. In words: $\mathbf{B}$ is orthogonal exactly when $\mathbf{A}$ is skew-symmetric.

Checks

- Take $\mathbf{A} = \begin{pmatrix} 0 & t \\ -t & 0 \end{pmatrix}$, which is skew-symmetric. Then $\det(\mathbf{I} - \mathbf{A}) = 1 + t^2 \neq 0$ and

$$\mathbf{B} = \frac{1}{1+t^2}\begin{pmatrix} 1 - t^2 & 2t \\ -2t & 1 - t^2 \end{pmatrix},$$

  a rotation matrix (put $t = \tan\frac{\theta}{2}$), so $\mathbf{B}^{\mathrm{T}}\mathbf{B} = \mathbf{I}$ as expected.
- The 1×1 case: $\mathbf{A} = (a)$ with $a \neq 1$ gives $\mathbf{B} = \left(\frac{1+a}{1-a}\right)$, and $\mathbf{B}^2 = 1$ forces $1 + a = \pm(1 - a)$, i.e. $a = 0$ (the other sign is impossible). This matches $2a = 0$.
- We tested about 2000 random matrices of sizes 1 to 5 with exact rational arithmetic: every skew-symmetric $\mathbf{A}$ gave $\mathbf{B}^{\mathrm{T}}\mathbf{B} = \mathbf{I}$ exactly, and none of the others did.

Common slips

- Writing $\mathbf{B}^{\mathrm{T}} = (\mathbf{I}+\mathbf{A}^{\mathrm{T}})(\mathbf{I}-\mathbf{A}^{\mathrm{T}})^{-1}$ without reversing the order. (It happens to be true, because $\mathbf{I}+\mathbf{A}^{\mathrm{T}}$ and $(\mathbf{I}-\mathbf{A}^{\mathrm{T}})^{-1}$ commute, but that needs its own justification.)
- Expanding $(\mathbf{I}+\mathbf{A}^{\mathrm{T}})(\mathbf{I}+\mathbf{A})$ with $\mathbf{A}\mathbf{A}^{\mathrm{T}}$ instead of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$. Here it cancels either way, but only if both sides are expanded consistently.
- Proving only one direction. The question asks for "if and only if", so say why each step can be reversed.

Beyond STEP

The map $\mathbf{A} \mapsto (\mathbf{I}+\mathbf{A})(\mathbf{I}-\mathbf{A})^{-1}$ is the *Cayley transform*. It turns skew-symmetric matrices, which form a vector space, into orthogonal matrices, which form a group, giving a rational parametrisation of rotations. (One can show that $\mathbf{I}-\mathbf{A}$ is automatically invertible when $\mathbf{A}$ is real and skew-symmetric, and that the orthogonal matrices reached are exactly those without eigenvalue -1.) This is a first glimpse of how a Lie group is related to its Lie algebra. Strang's *Introduction to Linear Algebra* is a good place to read more about orthogonal and skew-symmetric matrices.

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